

Quick Check

Lesson 4: Differences Between Acids and Bases Using Commercial Indicators

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Acids and Bases | Lesson 4

Two partners disagree about what colour methyl orange will turn in baking soda solution. Before testing, what is the best way to settle the disagreement?

- A** Carry out the test and record the colour actually seen
- B** Ask the teacher to say which prediction is right
- C** Vote and go with the answer most students choose
- D** Let the student who is more confident decide

In Experiment 1, a student puts 2-3 drops of freshly squeezed lemon juice on a strip of blue litmus paper. What colour does the paper turn?

A Red

B Blue (no change)

C Pink

D Colourless

A group predicted that cow's milk would be strongly basic. Their litmus and indicator results do not match this prediction. What should the recorders do?

- A** Write down the colours exactly as seen, even if they differ from the prediction
- B** Leave the milk row blank because the result was unexpected
- C** Copy the results from another group's table
- D** Change the results in the table to match the prediction

Liquid soap solution turns red litmus paper blue and turns phenolphthalein pink. How should it be classified, and why?

- A** Acid, because it changes the colour of litmus paper
- B** Base, because red litmus turns blue and phenolphthalein turns pink
- C** Neutral, because it does not make methyl orange red
- D** Acid, because phenolphthalein changes colour

A student tests tap water. Litmus paper shows no colour change, phenolphthalein stays colourless and methyl orange is orange-yellow. What is the best conclusion?

- A** It is neutral, because none of the indicators show an acid or base colour
- B** It cannot be classified because no indicator turned red
- C** It is a base, because methyl orange is yellowish
- D** It is an acid, because phenolphthalein is colourless

After Experiment 1, students can tell whether a liquid is an acid, a base or neutral using indicators. Which question is still NOT answered by this experiment?

- A** How do acids react, for example do they corrode or react with other substances?
- B** Is baking soda solution a base?
- C** What colour does phenolphthalein turn in a base?
- D** Is vinegar (siki) an acid?

A student revises a model by labelling the liquids tested in Experiment 1 using the evidence. Which set of labels is correct?

- A** Lemon juice: base; baking soda solution: acid; tap water: neutral
- B** Lemon juice: neutral; baking soda solution: base; tap water: acid
- C** Lemon juice: acid; baking soda solution: neutral; tap water: base
- D** Lemon juice: acid; baking soda solution: base; tap water: neutral

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).